

Windows Logging for SOC — Incident Investigation Report

Prepared By: Majd Mashaleh

Investigation Type :Windows Log Analysis / Threat Hunting / DFIR

Platform :Windows

Lab :TryHackMe — Windows Logging for SOC

Date :May 2026

Executive Summary

This report documents the investigation of suspicious activity identified on a Windows workstation during the Windows Logging for SOC laboratory. The objective of the investigation was to analyze Windows Security logs, Sysmon telemetry, and PowerShell artifacts to identify indicators of compromise, reconstruct attacker behavior, and determine the overall security impact.

The investigation identified evidence of:

- Repeated failed authentication attempts
- Successful malicious RDP authentication
- Unauthorized account creation
- Privilege escalation activity
- Suspicious process execution
- Malware download activity
- Persistence mechanisms
- Command-and-control communication
- Malicious PowerShell usage

Correlation across multiple Windows log sources enabled reconstruction of the attacker timeline and identification of malicious behavior consistent with a full intrusion lifecycle

1. Investigation Objectives

The primary objectives of the investigation were:

- Detect unauthorized authentication attempts
- Investigate successful logon events
- Identify suspicious process execution
- Analyze Sysmon telemetry
- Detect privilege escalation behavior
- Investigate persistence mechanisms
- Reconstruct attacker timeline
- Identify indicators of compromise (IOCs)
- Recommend mitigation and detection improvements

2. Log Sources Investigated

2.1 Windows Security Logs

Authentication and account activity were analyzed using:

Event Viewer → Windows Logs → Security

Purpose

1. Failed logon detection
2. Successful authentication analysis
3. RDP login monitoring
4. User management investigation
5. Privileged group modifications

Key Event IDs

4624 — Successful Logon

4625 — Failed Logon

4720 — User Account Created

4732 — User Added to

Privileged Group

Screenshot Placement

Open the "Practice-Security.exe" file on the VM's Desktop.

Which IP performed a brute force of the THM-PC?

10.10.53.248

✓ Correct Answer 0

Which user has been breached as a result of the attack?

Administrator

✓ Correct Answer 0

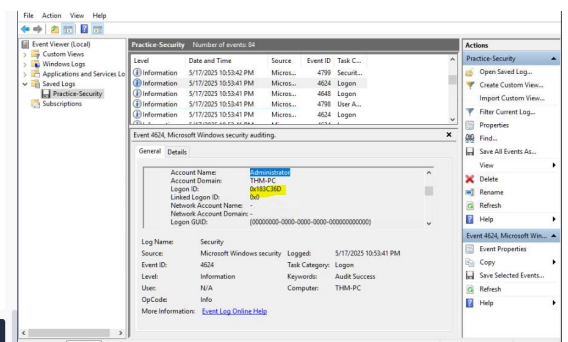
What was the Logon ID of the malicious RDP login?

Note: The login you are looking for has a Logon Type 10.

0x183C36D

✓ Correct Answer 0

Task 4 Security Log: User Management



2.2 Sysmon Operational Logs

Sysmon telemetry was analyzed using:

Applications and Services Logs → Microsoft → Windows → Sysmon → Operational

Purpose

1. Process creation monitoring
2. Network connection tracking
3. File creation monitoring
4. DNS request analysis
5. Malware execution detection

Key Event IDs

Event ID 1 — Process Creation

Event ID 3 — Network Connection

Event ID 11 — File Creation

Event ID 22 — DNS Query

Screenshot Placement

Answer the questions below

Continue with the "Practice-Sysmon.evtx" file on the VM's Desktop.

Which file was created by the downloaded malware to persist on the host?

C:\Users\sarah.miller\AppData\Roaming\Microsoft\W

✓ Correct Answer

What is the Command & Control server malware connected to?
(Answer in format IP:Port, e.g. 1.1.1.1:80)

193.46.217.4:7777

✓ Correct Answer

Finally, which domain does the malicious IP correspond to?

Level	Date and Time	Source	Event ID	Task C...
Information	5/18/2025 4:08:58 PM	Sysmon	10	Proces...
Information	5/18/2025 4:08:58 PM	Sysmon	10	Proces...
Information	5/18/2025 4:08:58 PM	Sysmon	3	Networ...
Information	5/18/2025 4:08:53 PM	Sysmon	22	Dns qu...
Information	5/18/2025 4:08:48 PM	Sysmon	22	Dns qu...

Event 3, Sysmon

General Details

SourcePort: 50538
SourcePortName: -
DestinationIP: false
DestinationIP: 193.46.217.4
DestinationHost: -
DestinationPort: 7777
DestinationPortName: -

Log Name: Microsoft-Windows-Sysmon/Operational
Source: Sysmon
Event ID: 3
Level: Information
User: SYSTEM
OpCode: Info

Logged: 5/18/2025 4:08:58 PM
Task Category: Network connection detected (rule:
Keywords:
Computer: THM-PC

More Information: [Event Log Online Help](#)

2.3 PowerShell History Analysis

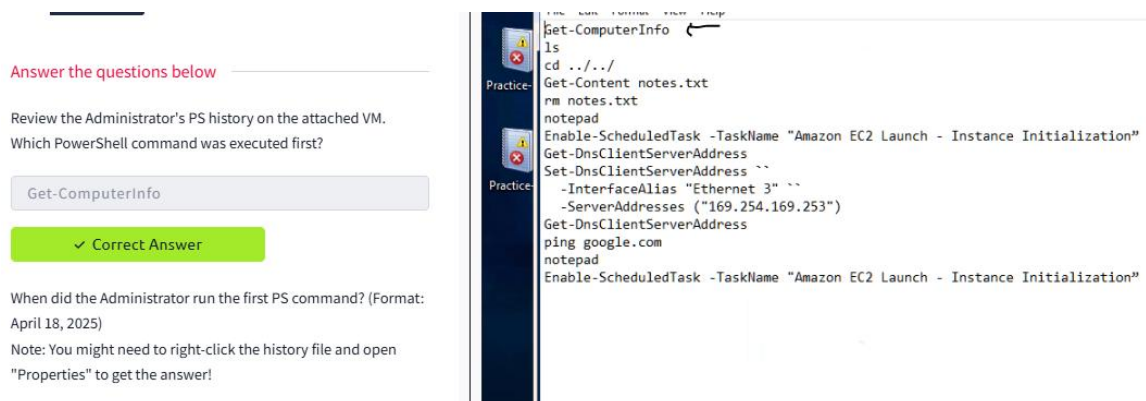
PowerShell activity was investigated using:

C:\Users\<USER>\AppData\Roaming\Microsoft\Windows\PowerShell\PSReadline\ConsoleHost_history.txt

Purpose

1. PowerShell command auditing
2. Malware download detection
3. Administrative command analysis
4. User activity investigation

Screenshot Placement



3. Investigation Methodology

The investigation followed a standard SOC triage and DFIR workflow:

1. Review Windows Security logs
2. Identify failed authentication attempts
3. Investigate successful RDP logons
4. Correlate suspicious Logon IDs
5. Analyze account management activity
6. Review Sysmon process execution logs
7. Investigate network communication events
8. Analyze PowerShell history artifacts
9. Reconstruct attacker timeline
10. Document findings and evidence

4. Authentication Analysis

4.1 Failed Authentication Attempts

The following filtering activity was performed:

Event ID: 4625

Findings

Multiple failed login attempts were identified targeting the Windows host through Remote Desktop Protocol (RDP). The attack originated from a suspicious external IP address.

Analysis

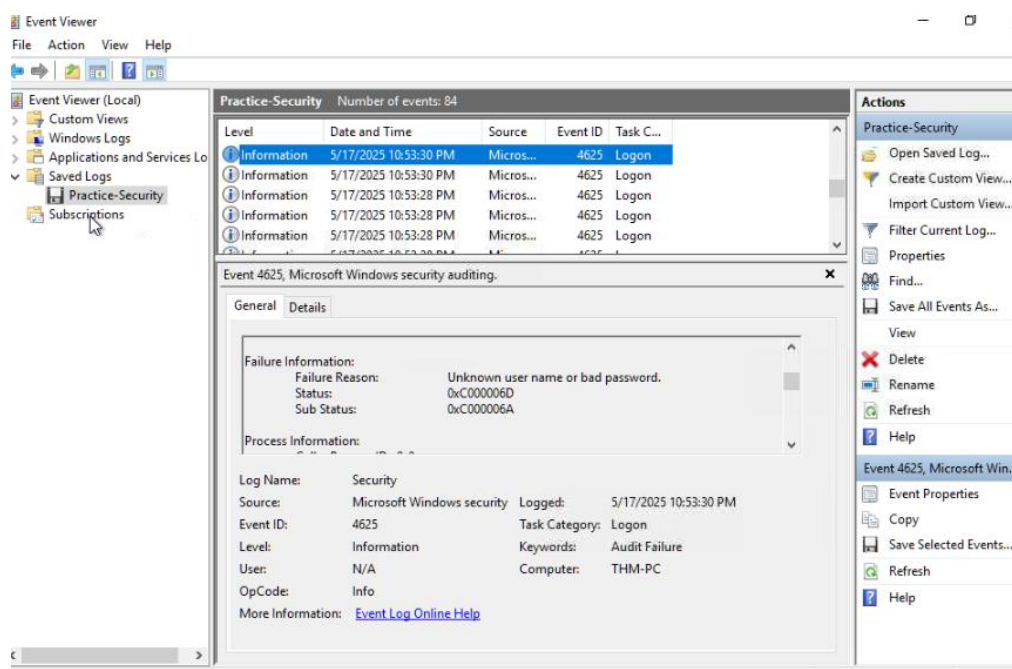
Observed behavior is consistent with:

- RDP brute-force attacks
- Password spraying
- Credential stuffing attempts

Security Impact

Repeated failed authentication attempts indicate active adversary probing and unauthorized access attempts against the Windows workstation.

Screenshot Placement



4.2 Successful Malicious RDP Authentication

The following filter was applied:

Event ID: 4624

Findings

A successful RDP login was identified shortly after the failed authentication attempts. Logon Type 10 confirmed remote interactive access through RDP.

Analysis

The authentication sequence strongly suggests:

- Credential compromise
- Successful attacker access
- Initial foothold establishment

Security Impact

Successful malicious RDP authentication significantly increases the probability of full host compromise.

Screenshot Placement

The screenshot displays a security training interface on the left and a Windows Event Viewer window on the right.

Security Training Interface (Left):

- Active machines information:** IP Address: 10.113.190.246
- Task description:** For this task, open the VM and analyze the logs of the RDP breach scenario: C:\Users\Administrator\Desktop\Practice\RDP Case\RDP-Security.evtx
- Question 1:** Which user seems to be most actively brute-forced by botnets?
Answer: Administrator (Correct Answer)
- Question 2:** Which IP managed to breach the host via RDP (Logon Type 10)?
- Check button:** A button labeled "Check" is visible.

Windows Event Viewer (Right):

- Filter:** RDP-Security (Number of events: 1,567)
- Event List:**

Level	Date and Time	Source	Event ID	Task Category
Information	5/20/2025 9:52:01 AM	Microsoft W...	4625	Logon
Information	5/20/2025 9:51:58 AM	Microsoft W...	4625	Logon
Information	5/20/2025 9:51:48 AM	Microsoft W...	4624	Logon
Information	5/20/2025 9:51:48 AM	Microsoft W...	4624	Logon

- Event Details (Event 4624, Microsoft Windows security auditing):**

General	
Account Domain:	WORKGROUP
Logon ID:	0x3E7
Logon Information:	
Logon Type:	10
Restricted Admin Mode:	No
Virtual Account:	No
Log Name:	Security
Source:	Microsoft Windows security
Event ID:	4624
Level:	Information
User:	N/A
OpCode:	Info
More Information: Event Log Online Help	

- Taskbar:** The system clock shows 1:11 PM on 5/3/2025.

4.3 Logon ID Correlation

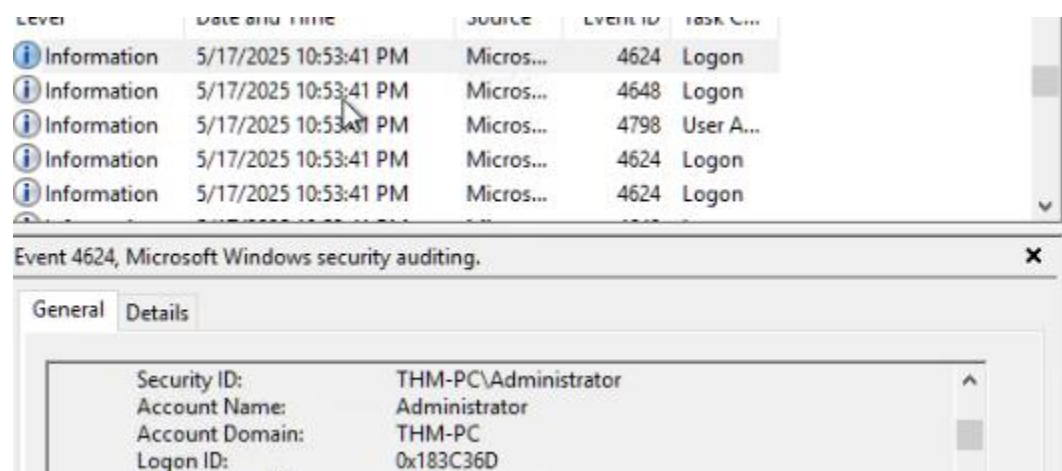
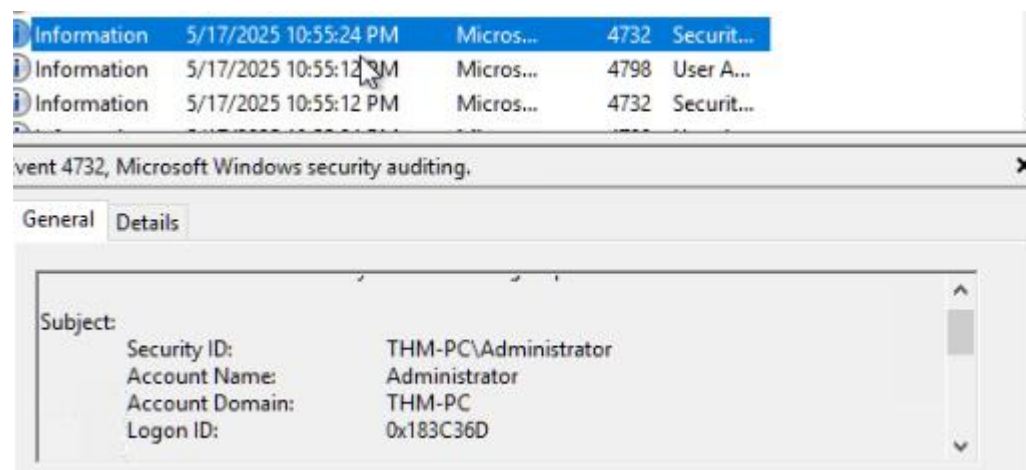
Findings

The malicious session was tracked using the Logon ID associated with the attacker's RDP session.

Analysis

Correlation of Logon IDs enabled tracking of:

- User management activity
- Process execution
- Persistence establishment
- Privilege escalation operations



5. User Management Investigation

5.1 Unauthorized Account Creation

The following Event ID was investigated:

4720 — User Account Created

Findings

A suspicious user account was created shortly after the successful RDP login.

Analysis

1. Observed behavior indicates:
2. Persistence establishment
3. Backdoor account creation
4. Long-term unauthorized access preparation

Security Impact

Unauthorized account creation allows attackers to maintain persistent access even after password resets.

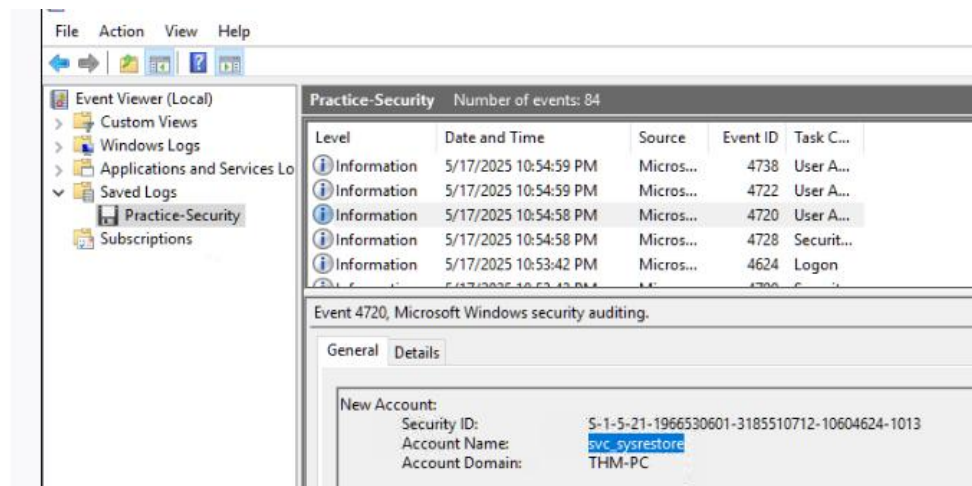
Screenshot Placement

Answer the questions below

Continue with the "**Practice-Security.evtx**" file on the VM's Desktop.

Which user was created by the attacker soon after the RDP login?

✓ Correct Answer



5.2 Privileged Group Membership Modification

The following Event ID was analyzed:

4732 — Member Added to Security-Enabled Local Group

Findings

The newly created account was added to privileged local groups.

Analysis

This activity indicates:

- Privilege escalation
- Administrative rights acquisition
- Persistence hardening

Security Impact

Attackers with administrative group membership gain full control over the compromised system.

Screenshot Placement

✓ Correct Answer

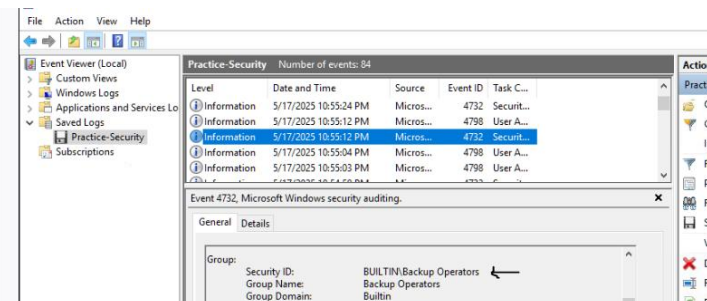
Which two privileged groups was the backdoor user added to?
(Answer in alphabetical order, e.g. "Administrators, Power Users")

Backup Operators, Remote Desktop Users

✓ Correct Answer

Does the Logon ID field match what you saw in the previous task (Yea/Nay)?

Yea



✓ Correct Answer

Which two privileged groups was the backdoor user added to?
(Answer in alphabetical order, e.g. "Administrators, Power Users")

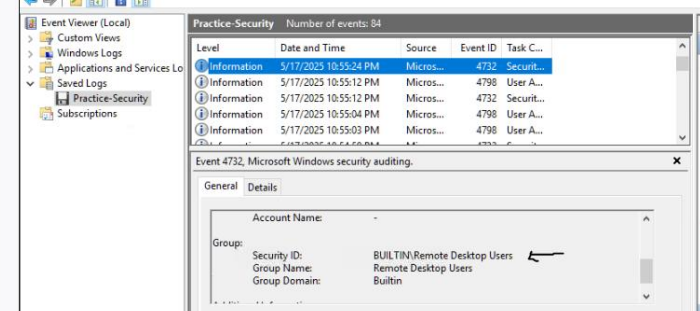
Backup Operators, Remote Desktop Users

✓ Correct Answer

Does the Logon ID field match what you saw in the previous task (Yea/Nay)?

Yea

✓ Correct Answer



6. Sysmon Investigation

6.1 Process Creation Analysis

The following Sysmon event was analyzed:

Sysmon Event ID 1 — Process Creation

Findings

Suspicious process execution activity involving browser downloads and malware execution was identified.

Analysis

Observed behavior suggests:

- Malware download activity
- Payload execution
- User-driven compromise chain

Security Impact

Malicious process execution may allow attackers to deploy persistence mechanisms and remote access malware.

Screenshot Placement

The image is a composite of two screenshots. The left screenshot shows a quiz interface with three questions and their correct answers. The right screenshot shows a Windows Event Viewer window displaying a Sysmon event.

Quiz Questions and Answers:

- Question: In which folder did the user unarchive the suspicious file?
Answer: C:\Users\Administrator\Downloads
- Question: What is the process ID of the launched phishing malware?
Answer: 5484
- Question: Finally, which malicious domain did the malware try to connect to?
Answer: (Not explicitly shown, but implied by the context of the quiz)

Event Viewer Details:

Level	Date and Time	Source	Event ID	Task Category
Information	5/20/2025 6:59:06 PM	Sysmon	1	Process Create
Information	5/20/2025 6:58:57 PM	Sysmon	1	Process Create
Information	5/20/2025 6:58:55 PM	Sysmon	1	Process Create
Information	5/20/2025 6:58:49 PM	Sysmon	1	Process Create

Event 1, Sysmon Details:

- Log Name: Microsoft-Windows-Sysmon/Operational
- Source: Sysmon
- Event ID: 1
- Level: Information
- User: SYSTEM
- OpCode: Info
- More Information: [Event Log Online Help](#)
- RuleName: -
- UtcTime: 2025-05-20 18:59:06.910
- ProcessGuid: {c5d2b969-d0fa-682c-2b05-000000001801}
- ProcessId: 5484
- Image: C:\Users\Administrator\Pictures\best-cat.jpg.exe
- FileVersion: -

6.2 File Creation Investigation

The following Sysmon event was reviewed:

Sysmon Event ID 11

Findings

File creation activity was observed during Sysmon analysis.

Analysis

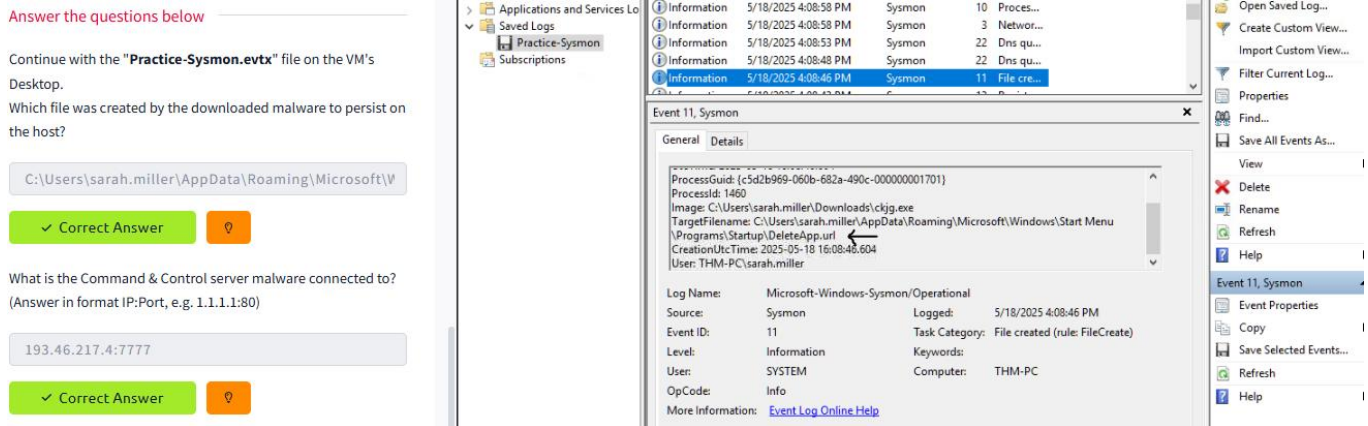
The event provided visibility into:

- File paths
- Creation timestamps
- Associated processes

Security Impact

File creation monitoring assists with identifying newly introduced files and suspicious system activity.

Screenshot Placement



6.3 Network Communication Analysis

The following event was analyzed:

Sysmon Event ID 3 — Network Connection

Findings

Outbound network connections to suspicious infrastructure were identified.

Analysis

Observed behavior indicates:

- Command-and-control communication
- Malware beaconing
- External attacker interaction

Security Impact

Command-and-control traffic confirms active malware communication with attacker infrastructure.

Screenshot Placement

The screenshot shows a CTF challenge interface on the left and a Windows Event Viewer window on the right.

CTF Challenge Interface:

- Path: `C:\Users\sarah.miller\AppData\Roaming\Microsoft\W`
- Question 1: "What is the Command & Control server malware connected to? (Answer in format IP:Port, e.g. 1.1.1.1:80)"
- Answer: `193.46.217.4:7777`
- Question 2: "Finally, which domain does the malicious IP correspond to?"
- Answer: `hkfasfsafg.click`
- Task 7: PowerShell: Logging Commands

Windows Event Viewer:

- View: Practice-Sysmon
- Number of events: 29
- Table of events:

Level	Date and Time	Source	Event ID
Information	5/18/2025 4:08:58 PM	Sysmon	11
Information	5/18/2025 4:08:58 PM	Sysmon	11
Information	5/18/2025 4:08:58 PM	Sysmon	21
Information	5/18/2025 4:08:58 PM	Sysmon	11
Information	5/18/2025 4:08:58 PM	Sysmon	11

Event 22, Sysmon Details:

- General: Dns query: RuleName: - UtcTime: 2025-05-18 16:08:57.208 ProcessGuid: {c5d2b969-060b-682a-490c-000000001701} ProcessId: 1460 QueryName: `hkfasfsafg.click` QueryStatus: 0
- Log Name: Microsoft-Windows-Sysmon/Operational
- Source: Sysmon
- Event ID: 22
- Level: Information
- User: SYSTEM
- OpCode: Info
- More Information: [Event Log Online Help](#)

7. PowerShell Investigation

7.1 PowerShell History Analysis

The following artifact was investigated:

`ConsoleHost_history.txt`

Findings

PowerShell history revealed administrative and reconnaissance-related commands executed on the compromised host.

Observed Commands

`Get-ComputerInfo`

`Get-LocalUser`

`Get-LocalGroup`

`Invoke-WebRequest`

Analysis

The commands indicate:

- System reconnaissance
- User enumeration
- Malware download activity

- Administrative discovery operations

Security Impact

PowerShell abuse remains one of the most common attacker techniques in Windows intrusions.

Screenshot Placement

Answer the questions below

Review the Administrator's PS history on the attached VM.
Which PowerShell command was executed first?

Get-ComputerInfo

✓ Correct Answer

When did the Administrator run the first PS command? (Format: April 18, 2025)

Note: You might need to right-click the history file and open "Properties" to get the answer!

```
Get-ComputerInfo
ls
cd ../../
Get-Content notes.txt
rm notes.txt
notepad
Enable-ScheduledTask -TaskName "Amazon EC2 Launch - Instance Initialization"
Set-DnsClientServerAddress
Set-DnsClientServerAddress -InterfaceAlias "Ethernet 3" -ServerAddresses ("169.254.169.253")
Get-DnsClientServerAddress
ping google.com
notepad
Enable-ScheduledTask -TaskName "Amazon EC2 Launch - Instance Initialization"
```

7.2 PowerShell Artifact Timestamp Analysis

Findings

The PowerShell history file creation timestamp confirmed the approximate time of initial PowerShell activity.

Screenshot Placement

Answer the questions below

Review the Administrator's PS history on the attached VM.
Which PowerShell command was executed first?

Get-ComputerInfo

✓ Correct Answer

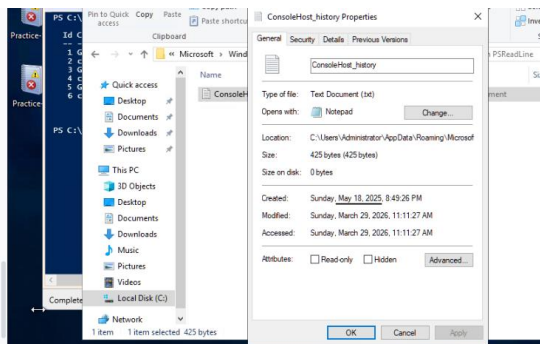
When did the Administrator run the first PS command? (Format: April 18, 2025)

Note: You might need to right-click the history file and open "Properties" to get the answer!

May 18, 2025

✓ Correct Answer

Can you find the flag stored in the PowerShell history? (Format: THM[...])



April 18, 2025)

Note: You might need to right-click the history file and open "Properties" to get the answer!

May 18, 2025

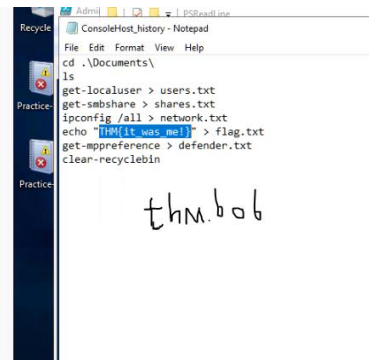
✓ Correct Answer

Can you find the flag stored in the PowerShell history? (Format: THM[...])

Note: You might want to check the PS history of other local users!

THM[it_was_me!]

✓ Correct Answer



9. Indicators of Compromise (IOCs)

Network Indicators

Suspicious External IP	Source of brute-force authentication activity
Malicious Domain	Command-and-control infrastructure

Process Indicators

powershell.exe	PowerShell execution
browser.exe	Malware download activity
suspicious executable	Malware payload

10. Behavioral Indicators

- Repeated RDP authentication failures
- Successful remote login after brute-force activity
- Unauthorized account creation
- Privileged group modification
- Suspicious process execution
- Malware download behavior
- PowerShell reconnaissance activity
- Command-and-control communication

12. Risk Assessment

Category	Severity
Unauthorized Access	High
Privilege Escalation	High
Persistence Risk	High
Malware Execution	High
Data Exposure	Medium
Command-and-Control Activity	High

13. Recommendations

- Immediate Actions
- Disable compromised accounts
- Reset exposed credentials
- Remove unauthorized users
- Block malicious IP addresses
- Isolate affected systems
- Conduct malware eradication procedures

14. Long-Term Improvements

- Enable Sysmon enterprise-wide
- Centralize logs into SIEM
- Configure MFA for RDP access
- Harden Windows authentication policies
- Monitor privileged group modifications
- Enable PowerShell logging controls
- Implement endpoint detection and response (EDR)

15. Detection Opportunities

Brute Force Detection

Monitor for:

Multiple Event ID 4625 failures from same source IP

Privileged Group Monitoring

Monitor for:

Event ID 4732 modifications involving Administrators group

PowerShell Abuse Detection

Monitor for:

- Invoke-WebRequest usage
- Encoded PowerShell commands
- Suspicious PowerShell history artifacts

16. Conclusion

The investigation identified multiple indicators of malicious activity on the Windows host.

Security log analysis revealed repeated brute-force authentication attempts followed by successful RDP compromise. Subsequent account creation and privileged group modification activity confirmed persistence establishment and privilege escalation.

Sysmon telemetry identified malware download activity, suspicious process execution, persistence mechanisms, and outbound command-and-control communication.

PowerShell history artifacts further confirmed attacker reconnaissance and operational activity on the host.

Correlation across Windows Security logs, Sysmon telemetry, and PowerShell artifacts enabled complete reconstruction of the attacker lifecycle and demonstrated the effectiveness of Windows logging in SOC investigations.

17. Analyst Notes

This investigation demonstrated practical experience in:

- Windows Event Log analysis
- SOC investigation methodology
- Threat hunting workflows
- Sysmon telemetry analysis
- Authentication monitoring
- PowerShell investigation
- Timeline reconstruction
- Incident reporting